Swat River and KPK flood dataset

Analysis notes on a synthetic hydro-meteorological dataset from Kaggle

What the dataset is

The download contains three CSV files about the Swat River basin in Khyber Pakhtunkhwa, Pakistan. The main one, daily_hydro_met.csv, holds 975 daily rows for a single gauge called Chakdara_Gauge, running from 1 January 2024 to 1 September 2026, with rainfall, average temperature, river discharge in cusecs, water level and a flood label. A second file lists one row per year from 2010 to 2026 with the annual peak discharge, a monsoon rainfall anomaly, a flood flag and damages. The third describes four sub-basins with elevation, slope, forest cover and an encroachment index.

The most important thing to know comes from the README shipped with the data: it describes this as a synthetic dataset, engineered from historical reports and typical discharge ranges for training flood-forecasting models. The daily series is therefore simulated, not observed. Nothing in this report should be read as a measurement of the real river. What follows describes how the generated data behaves and whether it is fit for the purpose the author suggests.

Cleaning

There was very little to repair, which is itself a sign of generated data. The daily file has no missing values, no duplicate rows, no duplicate dates and no gaps in the calendar across all 975 days. No value is negative, and the only station name is spelled the same way throughout.

The changes made were small. Dates were parsed from text into real dates. Column names were converted to lower-case snake_case with their units kept. Values stored to fifteen decimal places were rounded to a sensible precision (two decimals for rainfall and temperature, one for discharge, three for water level). Three helper columns were added: year, month, and a monsoon flag for July and August, the two months in which the simulated rainfall is high. The cleaned file has the same 975 rows and goes from 7 to 10 columns.

An interquartile-range check flagged 117 rainfall days and 122 discharge days as outliers. All 122 of the discharge days and 114 of the 117 rainfall days fall in July or August, so these are the simulated monsoon rather than errors, and nothing was removed. The station and flood label columns were kept even though both are constant.

What the daily series looks like

The year splits into two regimes. For ten months rainfall averages 1.98 mm a day; in July and August it averages 18.89 mm a day. Those two months make up 186 of the 975 days but carry 69.2 percent of all the rain in the file. The change happens abruptly on 1 July and reverses just as abruptly at the end of August. It also never stops raining: there is not one dry day in the whole series, and the smallest daily value is 0.01 mm.

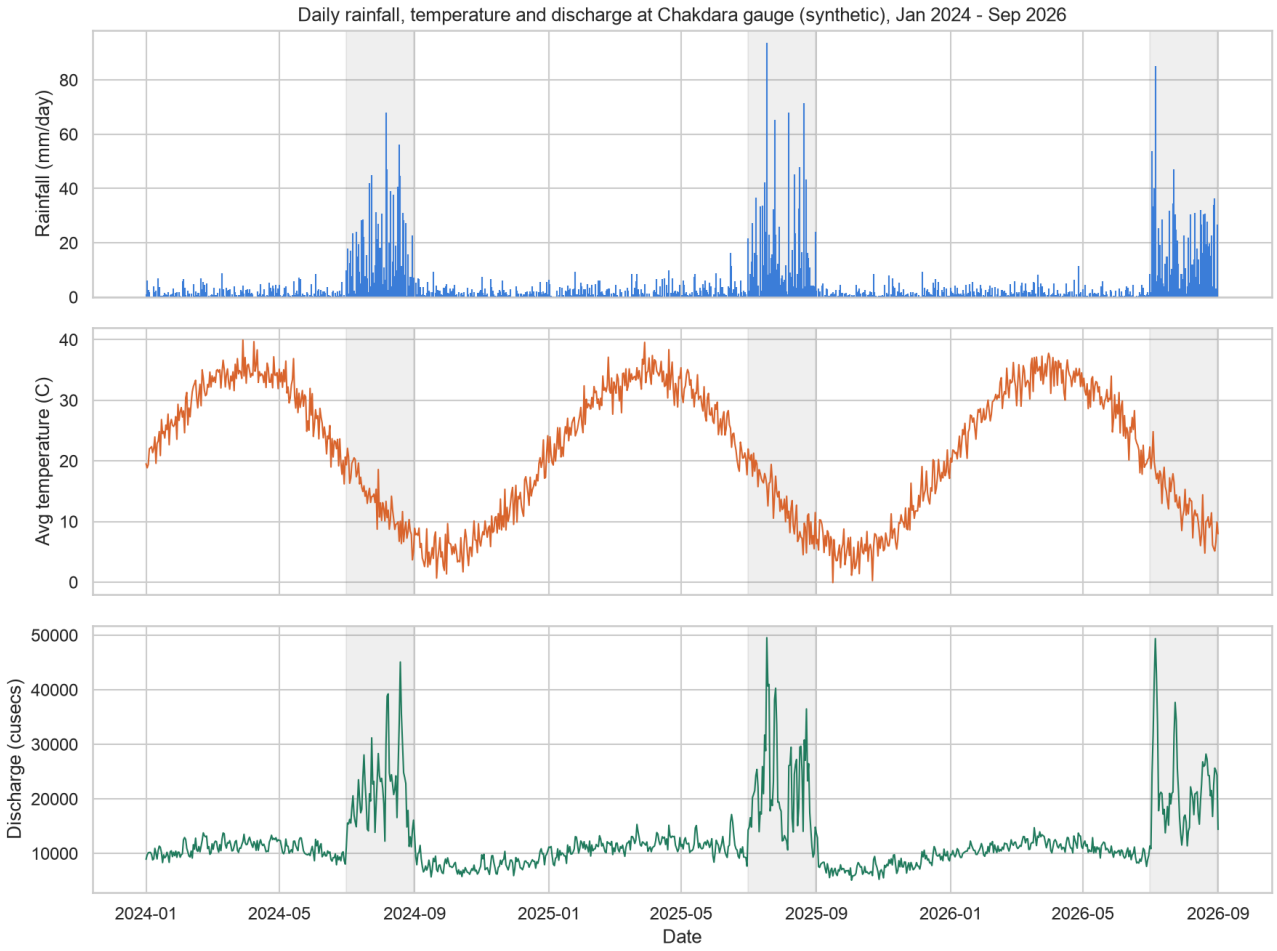


Figure 1. The three daily series. Grey bands mark July and August.

Discharge follows the rain. Mean discharge is 10,110 cusecs outside the monsoon and 21,840 cusecs within it, and the highest value on any non-monsoon day is 17,134 cusecs, so all eight days above 40,000 cusecs are in July or August. Outside the monsoon the series is a gentle seasonal wave with a little noise.

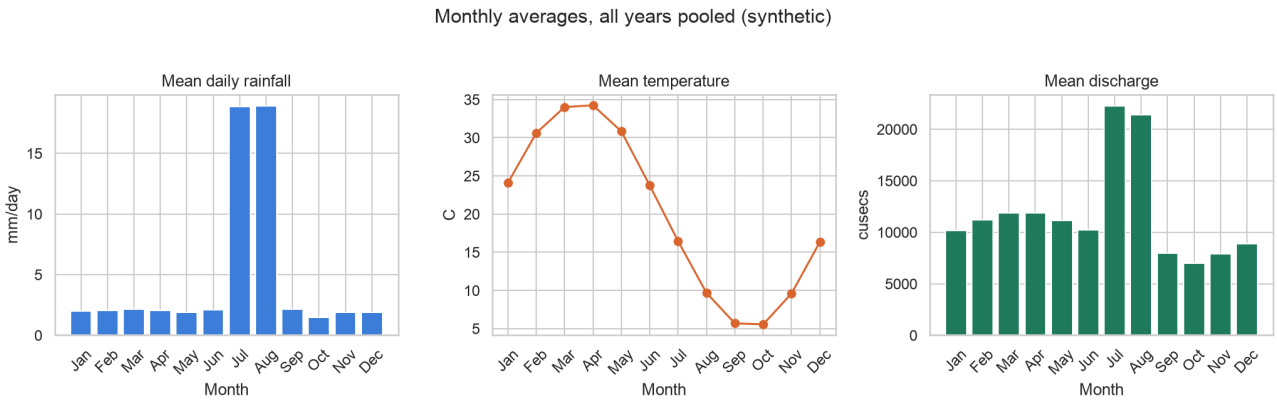


Figure 2. Monthly averages with all years pooled.

Temperature is where the simulation goes wrong. The monthly means peak in March and April (34.0 and 34.2 C) and bottom out in September and October (5.7 and 5.6 C). July averages 16.5 C and August 9.7 C. In the real Swat valley the hottest part of the year is early summer and the coldest is midwinter, so the cycle here appears to be shifted by roughly three months. That shift is what produces the mildly negative correlation between rainfall and temperature (-0.25). I would treat the temperature column as unreliable even by the standards of synthetic data.

Rainfall, discharge and water level

Same-day rainfall and discharge have a correlation of 0.77. Rainfall one and two days earlier correlates almost identically (0.76 and 0.77) and then the figure drops to about 0.52 at three days. The three-day rainfall total correlates with discharge at 0.96, higher than the two-day (0.90) or four-day (0.92) totals. That pattern suggests discharge was produced from a three-day rainfall window laid over a seasonal base flow, although the README does not document the formula. A model trained to predict discharge from rainfall would mostly be recovering that formula.

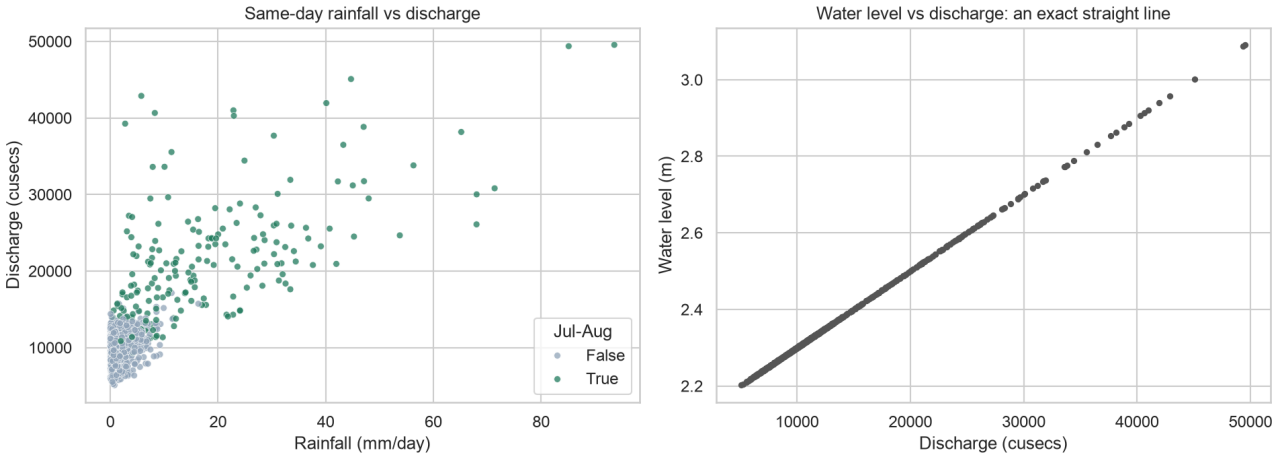


Figure 3. Rainfall against discharge (left) and discharge against water level (right).

Water level is not an independent measurement. It equals 2.1 m plus discharge divided by 50,000, with residuals at the level of floating-point error. Its correlation with discharge is exactly 1, so it should not be used as a feature alongside discharge or as evidence of anything.

The flood label deserves its own mention. The README proposes it as a classification target, but it is 0 on every one of the 975 days. There is no positive example to learn from.

The yearly flood table and how it relates to the daily data

The flood table has 17 rows, which is far too few for statistics, so this section only describes it. Four years carry the flood flag: 2010, 2020, 2022 and 2025. In this table the flag coincides exactly with a listed peak of 95,000 cusecs or more; the highest unflagged year is 2026 at 88,000. The 2022 row is the largest at 247,000 cusecs with a 78 percent rainfall anomaly and damages of PKR 35,000 million, followed by 2010 at 175,000 cusecs and PKR 12,000 million. Those two years account for 93.1 percent of the PKR 50,470 million of damages listed. Seven unflagged years still list some damages, and six years list none.

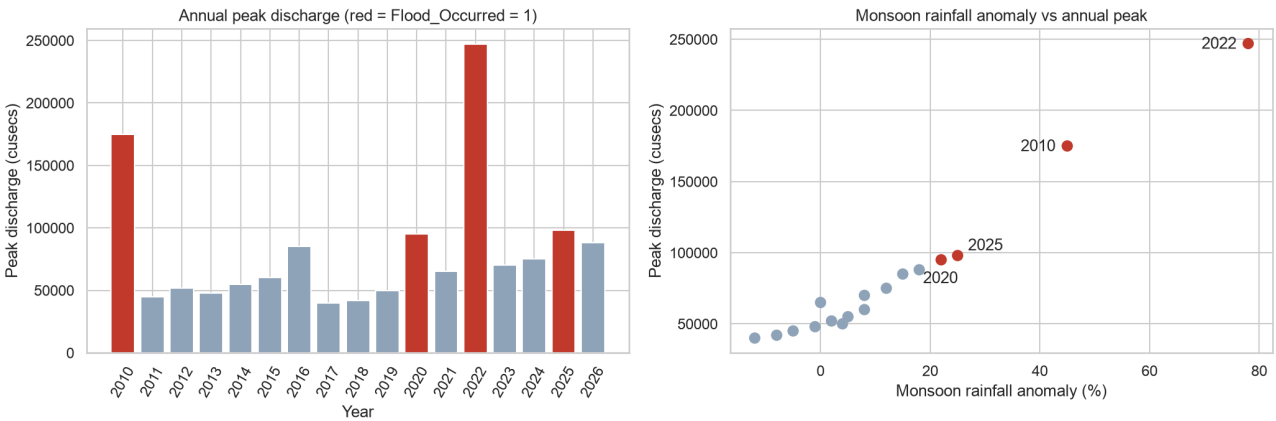


Figure 4. The yearly flood table. Red marks years with Flood_Occurred = 1.

The daily file and the flood table overlap for 2024, 2025 and 2026, so in principle the largest daily discharge in each year can be set against the listed annual peak. They do not match.

Year	Daily max (cusecs)	Date of daily max	Table peak (cusecs)	Daily max as % of table	Flood flag (table)	Flood-label days (daily file)
2024	45,078	19 Aug 2024	75,000	60.1%	0	0
2025	49,543	18 Jul 2025	98,000	50.6%	1	0
2026	49,372	6 Jul 2026	88,000	56.1%	0	0

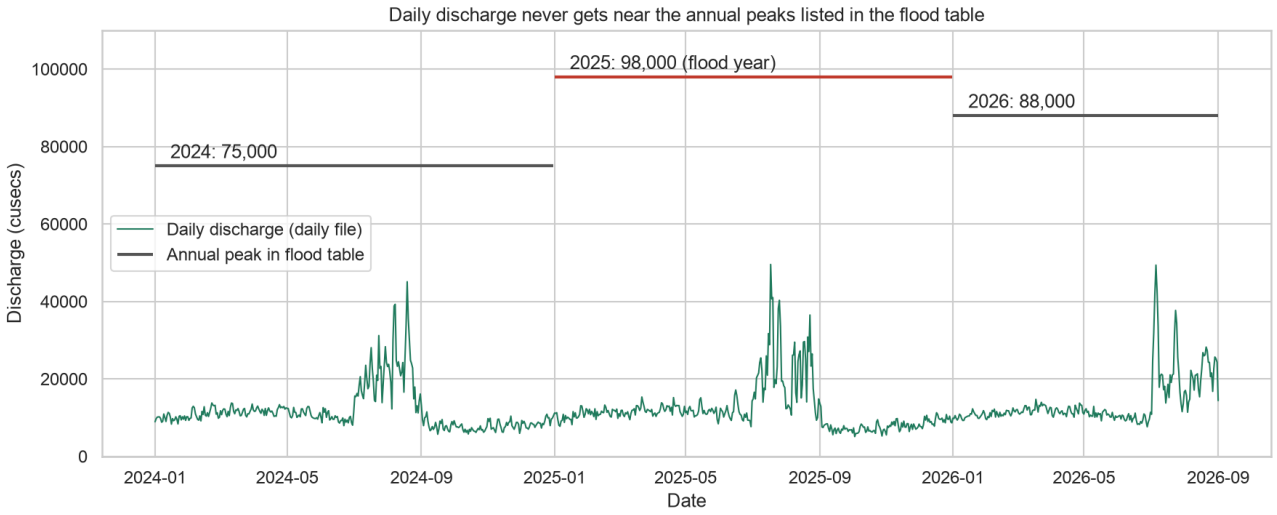


Figure 5. Daily discharge against the annual peaks listed for the same years.

The daily maximum reaches only 50 to 60 percent of the listed peak in each year, and no day in the daily file gets to 95,000 cusecs. The year 2025 is flagged as a flood in the table yet has no flood-labelled day. The table also lists monsoon rainfall anomalies of +12, +25 and +18 percent for the three years, while the July to August rainfall totals in the daily file (1,145, 1,178 and 1,190 mm) sit within about 2 percent of each other; the baseline for the anomaly is not documented, so that comparison is rough. The two files appear to have been generated independently, and the daily series cannot be used to examine the listed flood events.

Basin features

The basin table has four rows running upstream to downstream. Mean elevation falls from 2,500 m at Upper Swat (Kalam) to 680 m at Chakdara Downstream, average slope from 32 to 5 degrees and forest cover from 35 to 6 percent. The encroachment index is highest at Lower Swat (Mingora) with 9 out of 10, against 7 at Chakdara and 2 at Kalam. How that index was scored is not explained, and nothing links these rows to the daily series beyond the gauge name.

Highest, lowest, most

- Wettest day: 93.60 mm on 18 July 2025. Driest: 0.01 mm on 13 March 2024.
- Highest discharge: 49,542.5 cusecs on 18 July 2025, also the highest water level at 3.091 m.
- Lowest discharge: 5,089.7 cusecs on 3 October 2025 (water level 2.202 m).
- Largest one-day rise in discharge: 20,728 cusecs, on 18 July 2025.
- Hottest day: 39.93 C on 29 March 2024. Coldest: -0.07 C on 16 September 2025, the only day below zero.

- Wettest calendar month: August 2024 with 652.8 mm. Driest full month: June 2026 with 42.8 mm.
- Highest monthly mean discharge: July 2025 at 24,148 cusecs. Lowest: October 2025 at 6,868 cusecs.
- Days with 50 mm or more of rain: 8, all in July or August.
- Flood table: highest peak 247,000 cusecs (2022), lowest 40,000 cusecs (2017), a ratio of 6.18.
- Basin table: steepest and most forested is Upper Swat (Kalam); most encroached is Lower Swat (Mingora).

Conclusion and limitations

This is a small, tidy, simulated dataset. It works as a toy example of a monsoon-driven river, with a clear wet season and a discharge series that responds to rainfall in a simple way. It does not work for what the README proposes. The classification target is empty, the water level is a restatement of discharge, the temperature cycle is out of phase, and the daily series disagrees with the yearly flood table it is meant to accompany.

The limits are plain: one station, under three years, a partial final year, 17 yearly rows and 4 basin rows, and no documentation of how any column was generated. The flood table includes 2026 as a full-year entry even though the daily file stops on 1 September 2026. None of the figures here should be quoted as facts about real floods in Swat.

If someone wanted to take this further, the useful next step would be to replace the simulated series with observed gauge and rainfall records for the same basin and repeat the monthly and rainfall-response checks on those. Within this dataset, the only modelling exercise that makes sense is a discharge regression on lagged rainfall, with the understanding that it tests the generator and not the river.